HSDSLab
Human & Social Data Science

THE BRITISH JOURNAL OF PHARMACOLOGY

80 years of progress in Pharmacology

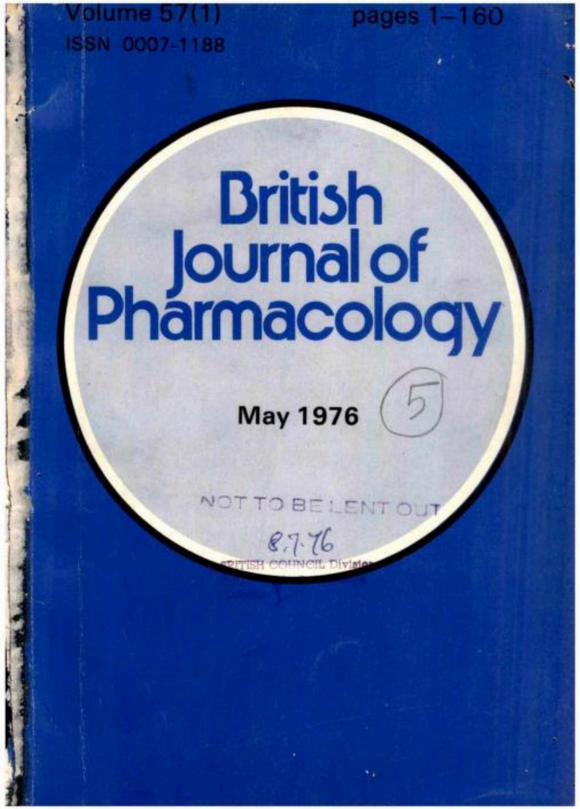
Gabriel Vignon – Under the supervision of Csaba Kiss and Dr. Roland Molontay

Contents

INTRODUCTION

Aims of the study

- Understanding **how Pharmacology has evolved** over the last 80 years, through the lens of the BJP publications.
- Study **relations between aggregated metadata** and **patterns in coauthorship and keywords networks** to improve understanding of knowledge diffusion in Pharmacology.
- No previous research on applying Science of Science specifically to Pharmacology.



Data

- OpenAlex, an open-source database, provides us with **26,280 works** published in the BJP.

Comparison with other databases:

- Web of Science: 35,117 works,
- Pubmed: 25,582 works

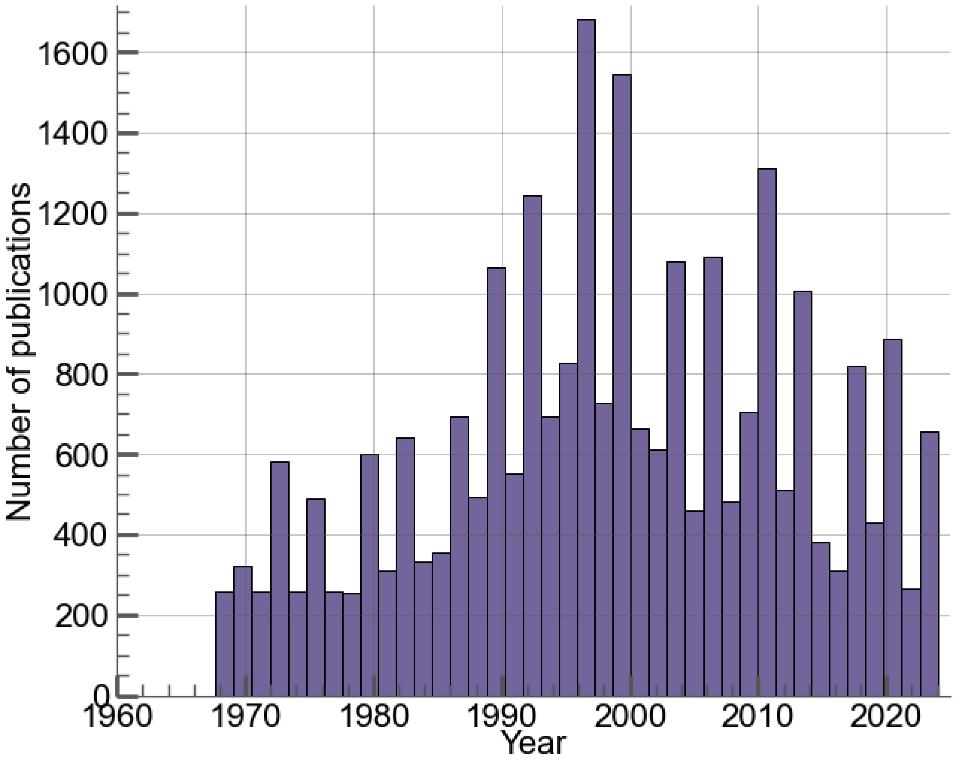


Figure 2: Number of works per year

MAIN RESULTS

Comparison with other pharmaceutical journals

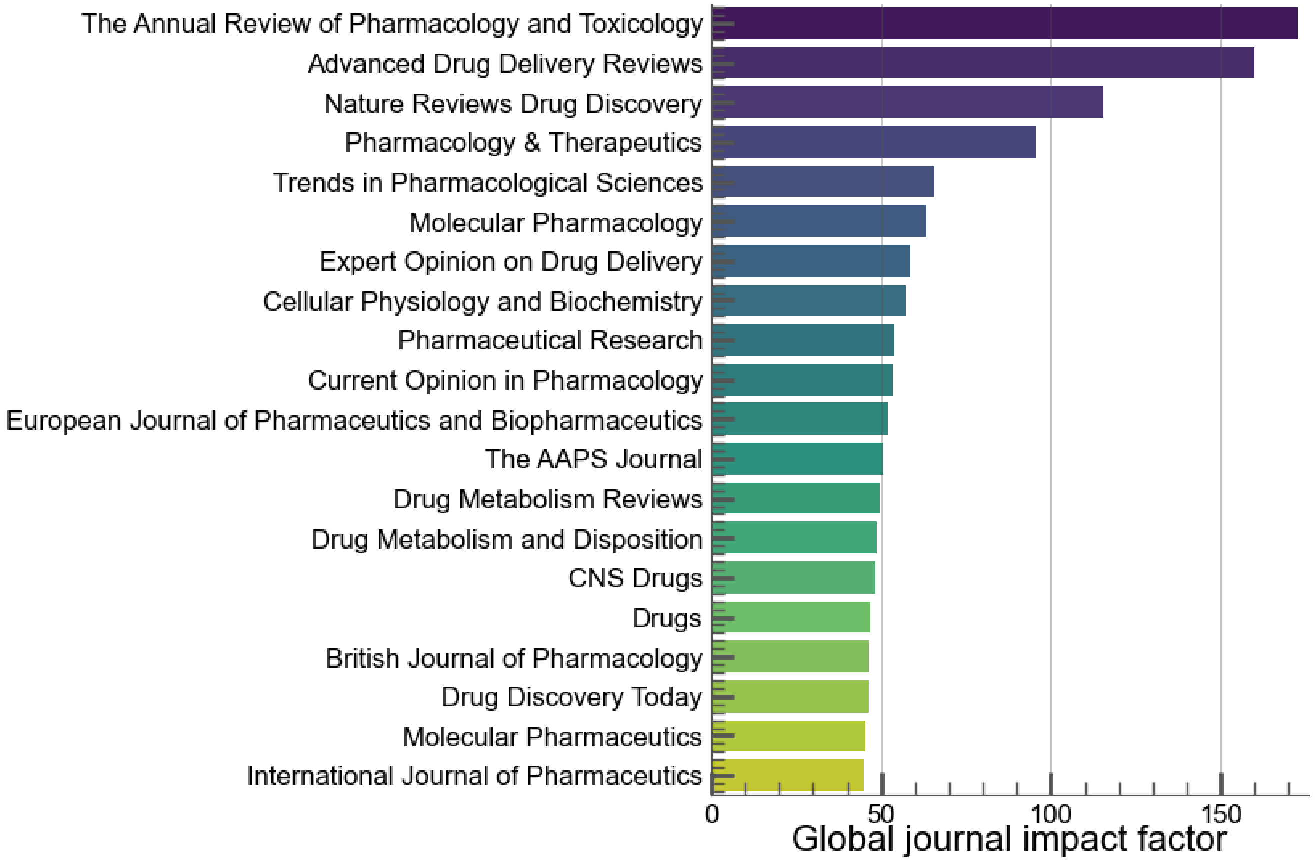


Figure 3: Top 20 journals in terms of number of citations per work

Top authors and countries

HSDSLab
Human & Social Data Science

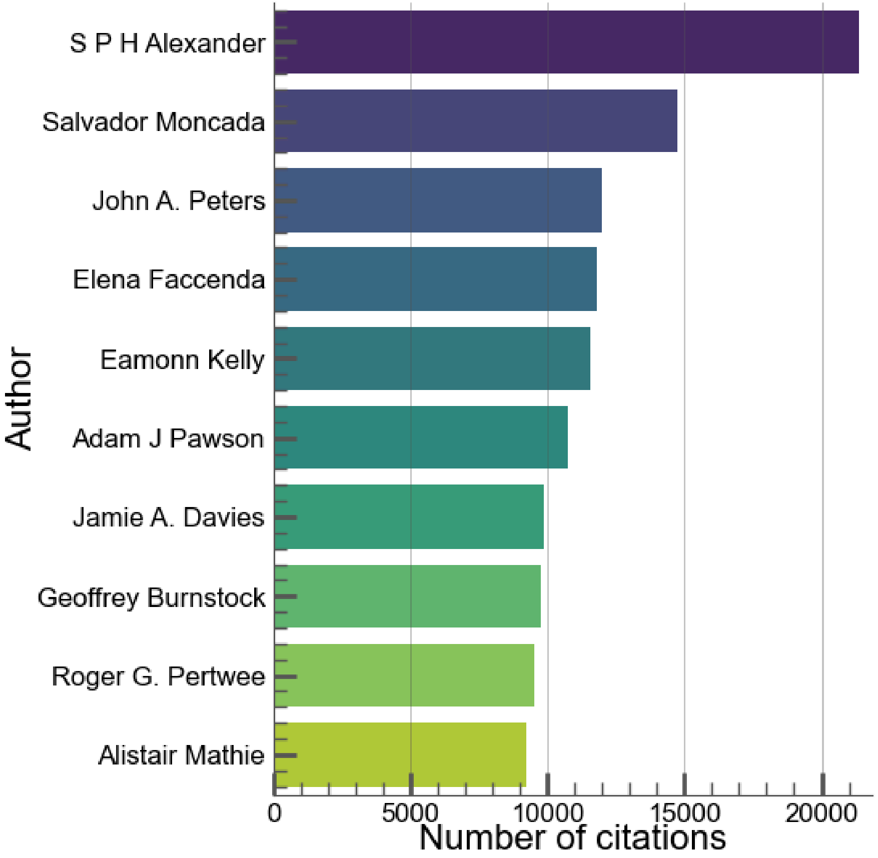


Figure 4: Top 10 authors by number of citations (regardless of authorship position)

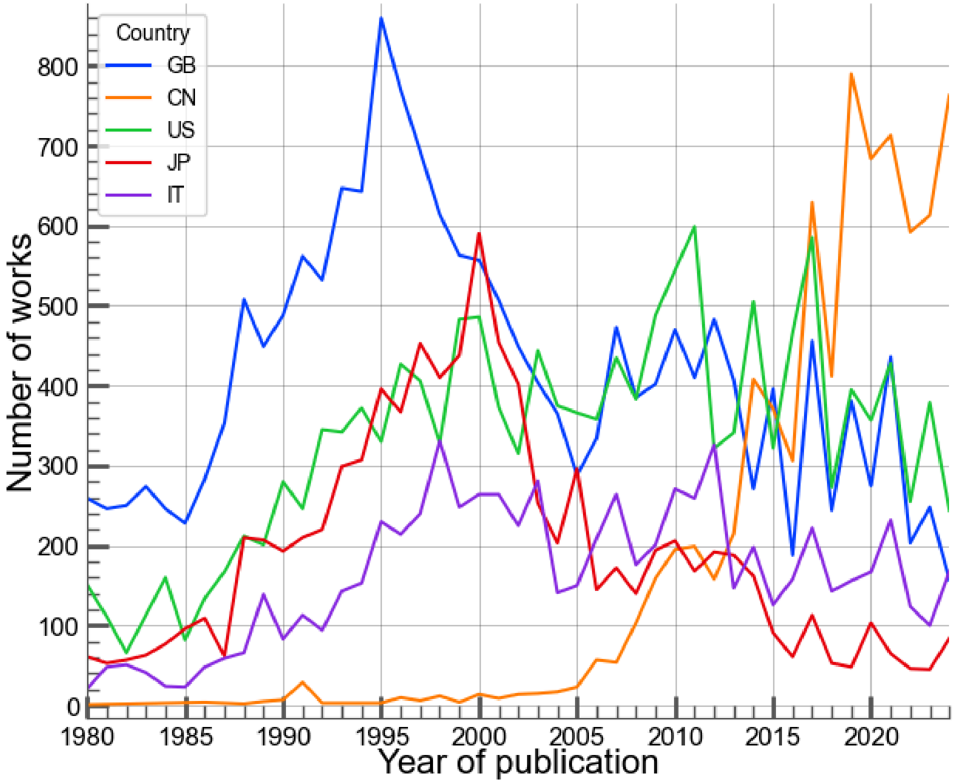


Figure 5: Top 5 countries in terms of number of works published in the BJP

Top institutions

HSDSLab
Human & Social Data Science

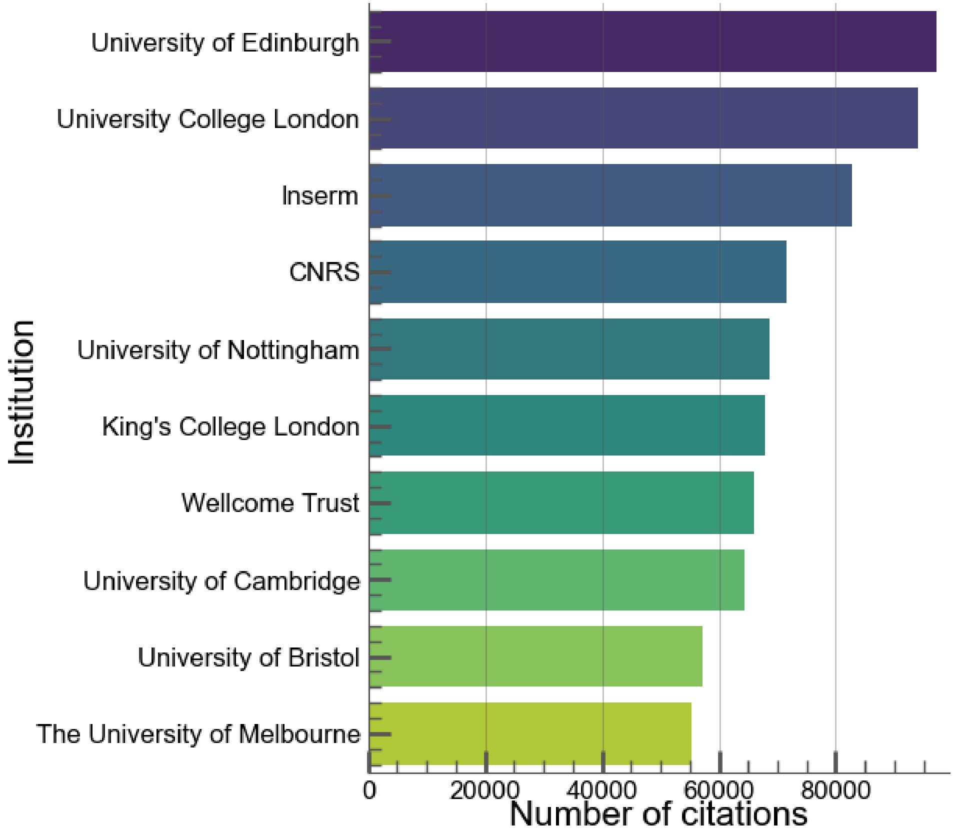


Figure 6: Top 10 institutions by number of citations (all kind of institutions)

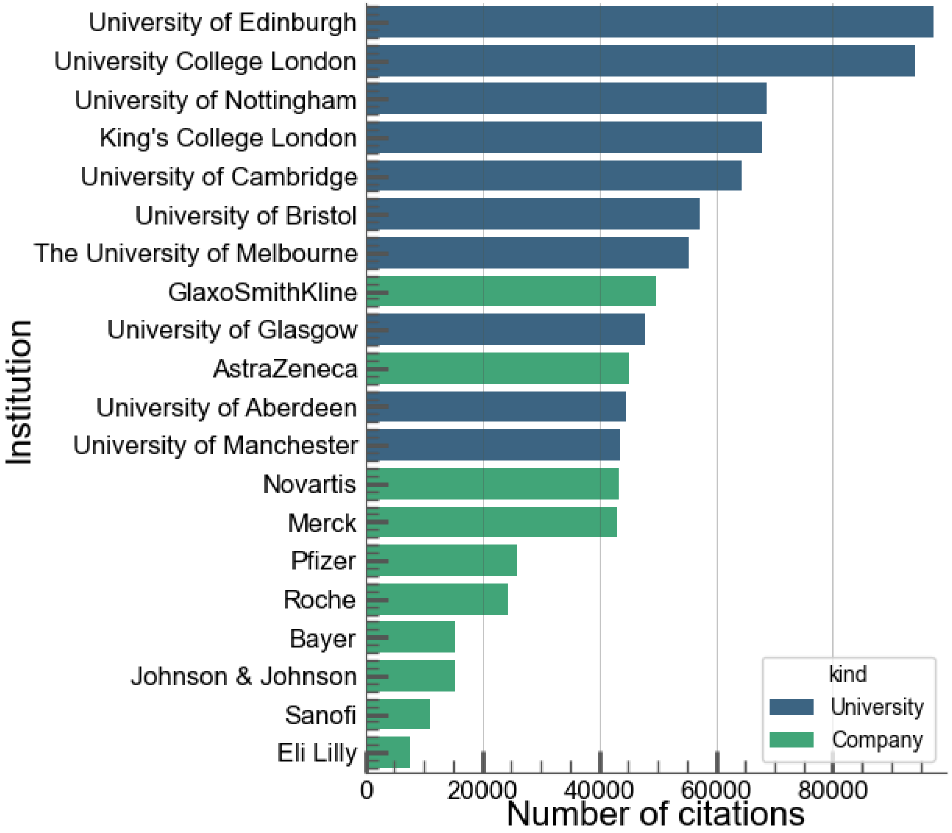


Figure 7: Top 10 universities (blue, 70 citations per work) versus top 10 companies (green, 72 citations per work)

Citations distribution

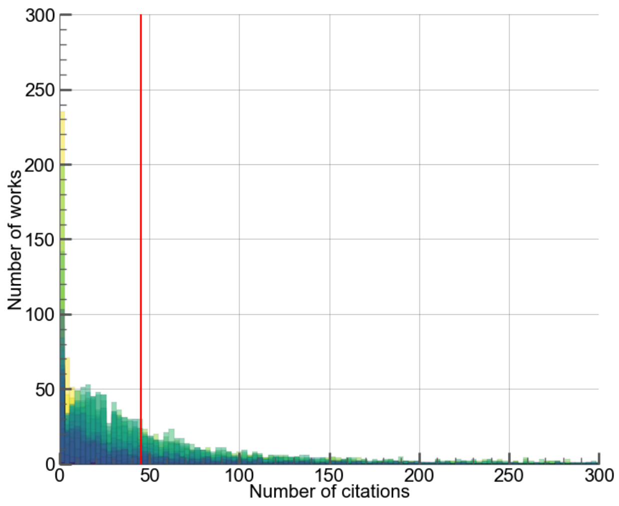


Figure 8: Distribution of the number of citations per work (fit with a Pareto distribution). Mean in red.

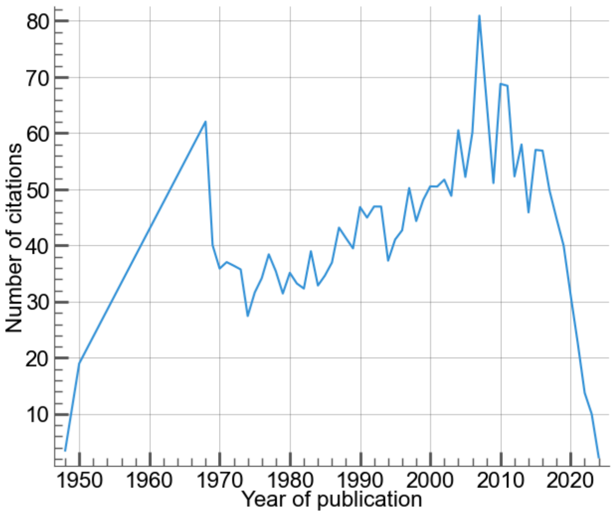


Figure 9: Mean number of citations per work according to their publication years

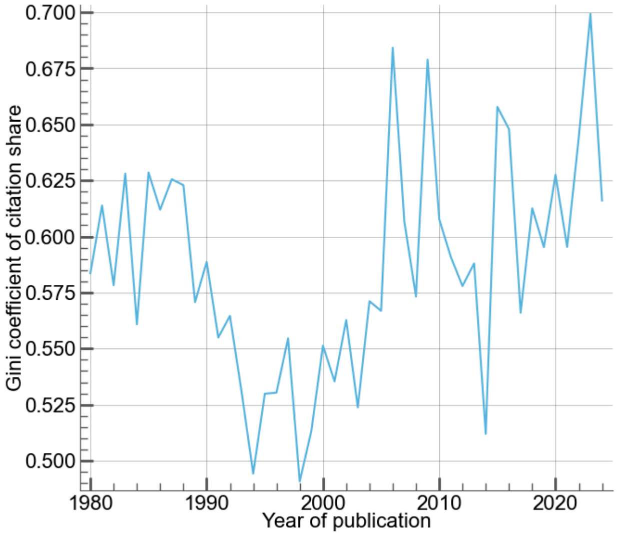


Figure 10: Citation share inequalities index between works (Gini coefficient).

S

Changes of practices over the years

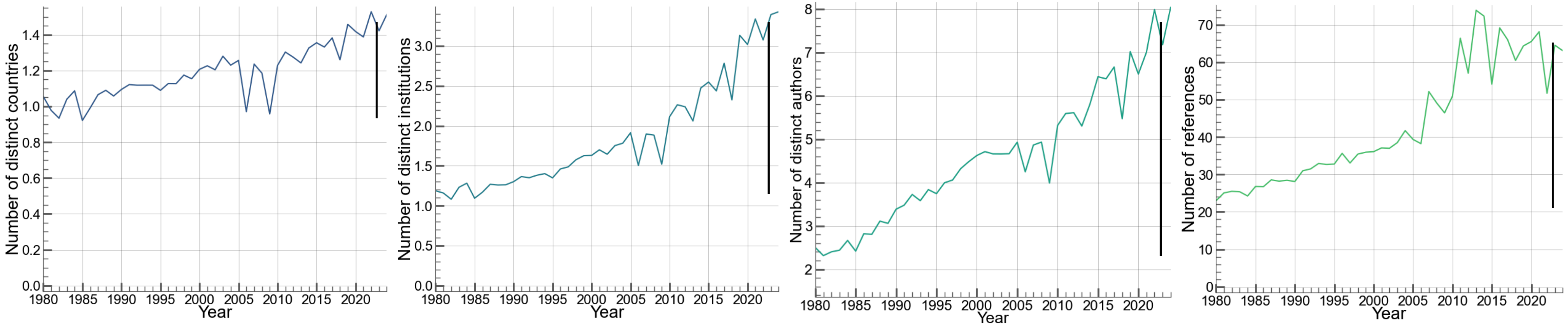


Figure 11: **Left:** Mean number of countries involved per work. **Center Left:** Mean number of institutions involved in a work. **Center Right:** Mean number of authors per work. **Right:** Mean number of references per work.

Countries comparison

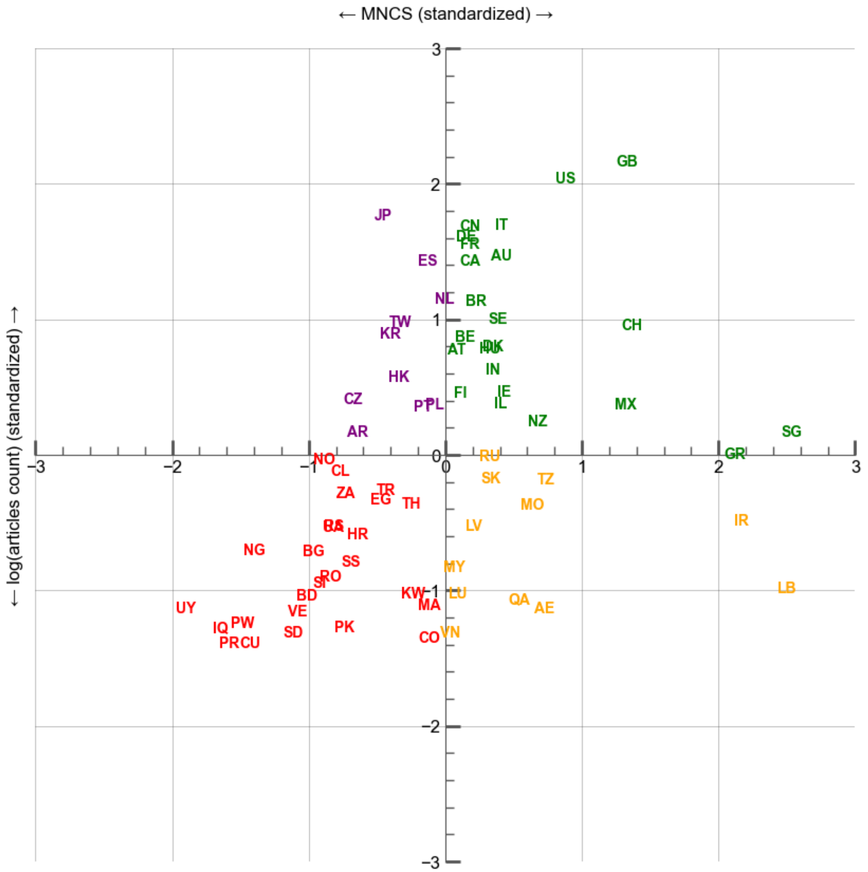


Figure 12: Countries statistics, MNCS

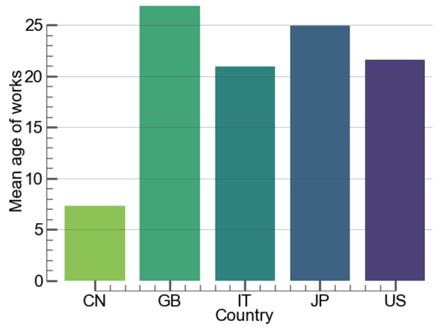


Figure 13: Mean age of articles, by country

Addressing the age bias:

$$\text{Mean Normalized Citation Score} = \frac{\text{citations of the work}}{\text{mean citations for the same year works}}$$

Correlations between impact (MNCS) and other factors

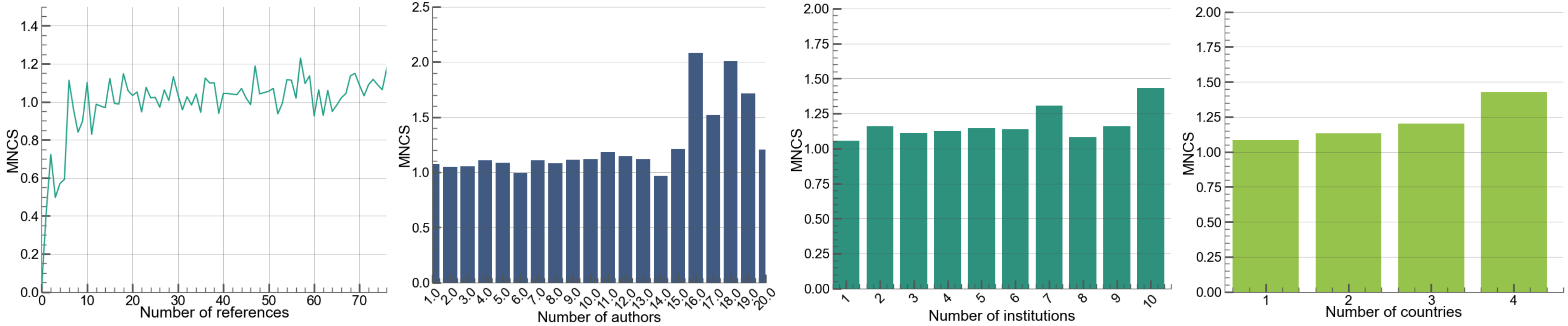


Figure 14: Mean number of citations, grouped by number of **references (left)**, of **authors (center left)**, of **institutions (center right)** and of **countries (right)**. On average, each work involves 4.0 authors, 1.7 institutions and 1.2 countries.

Correlations between impact (MNCS) and other factors

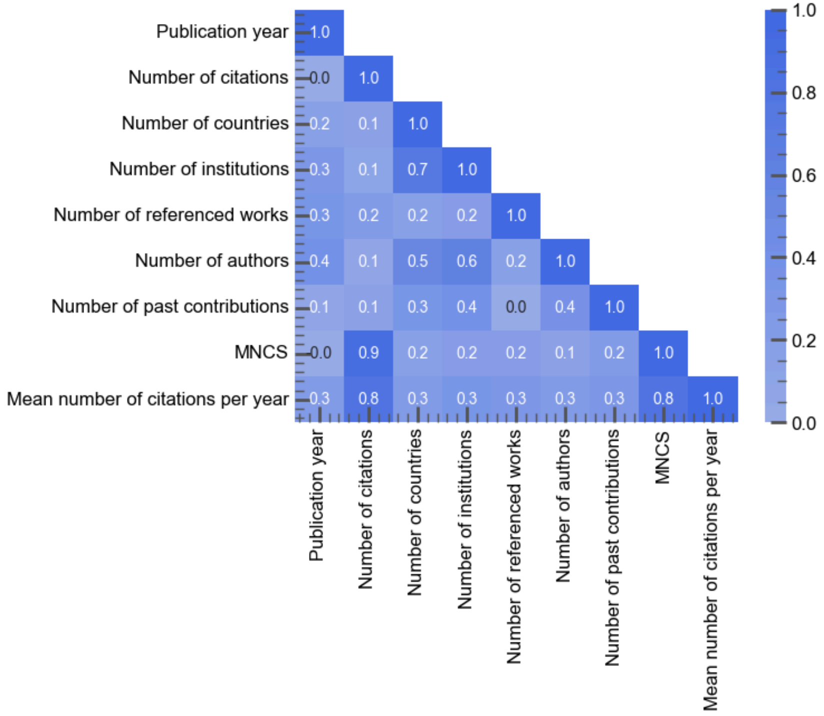


Figure 15: Heatmap of correlations

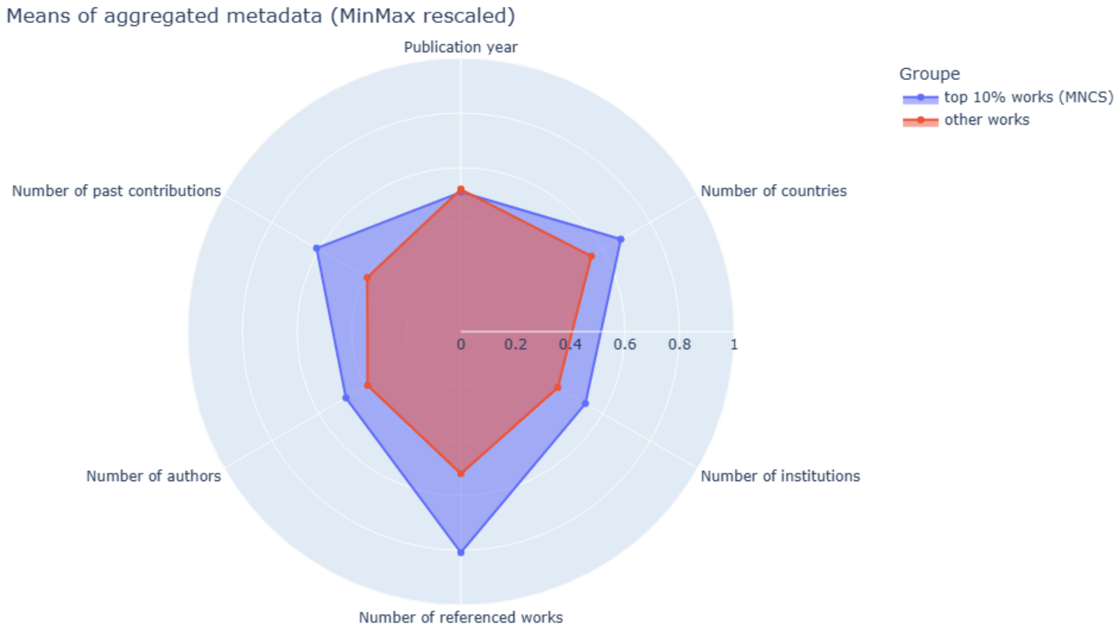


Figure 16: Mean factors of top 10% works versus other ones

Life of works

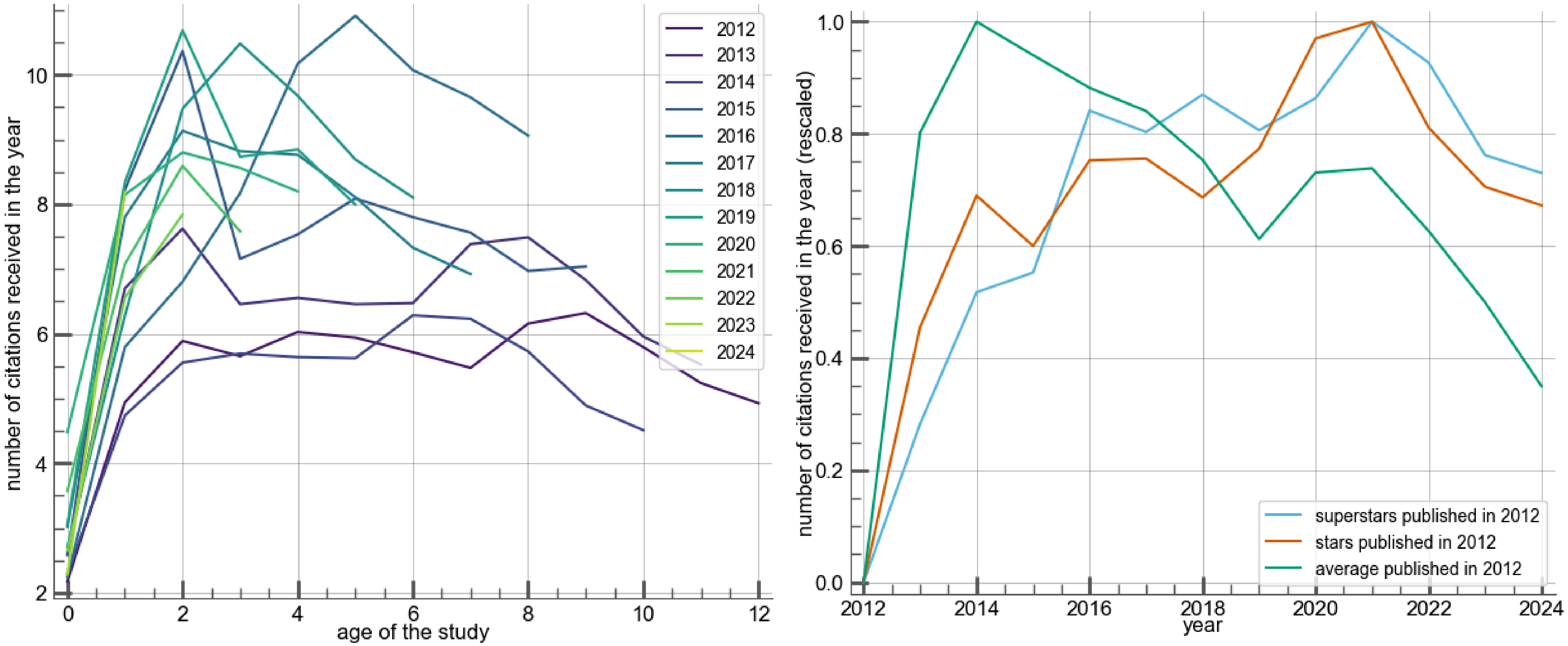


Figure 17: **Left:** Life of works (since 2012). **Center:** Beginning of the life of recent works according to their publication year. Year 0 corresponds to each work's publication date; data is grouped accordingly. **Right:**

Coauthorship networks

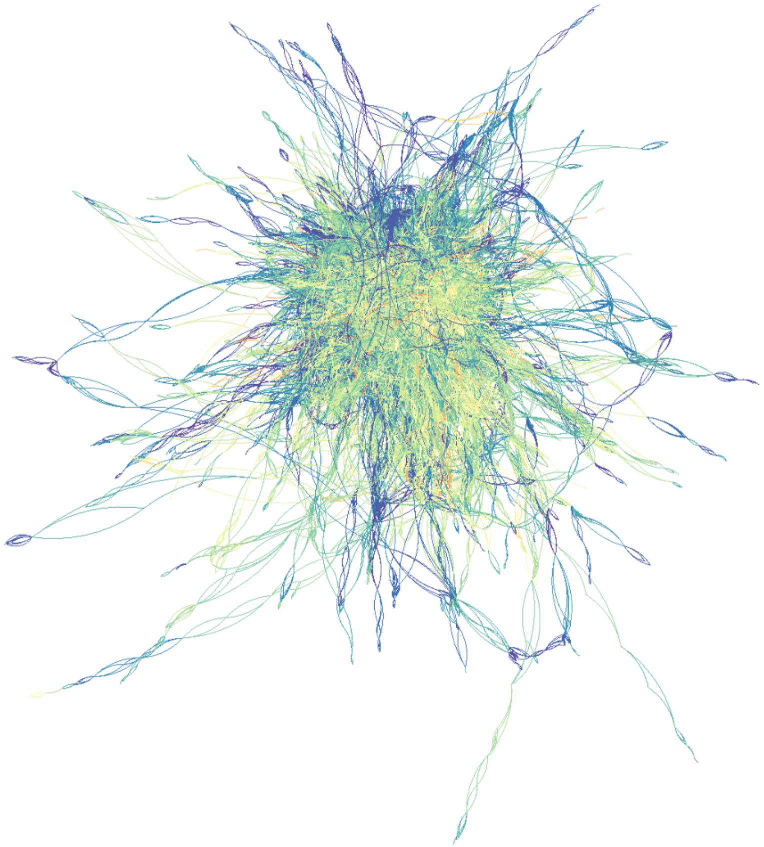


Figure 18: Main connected component of the graph - oldest edges in red, newest in purple

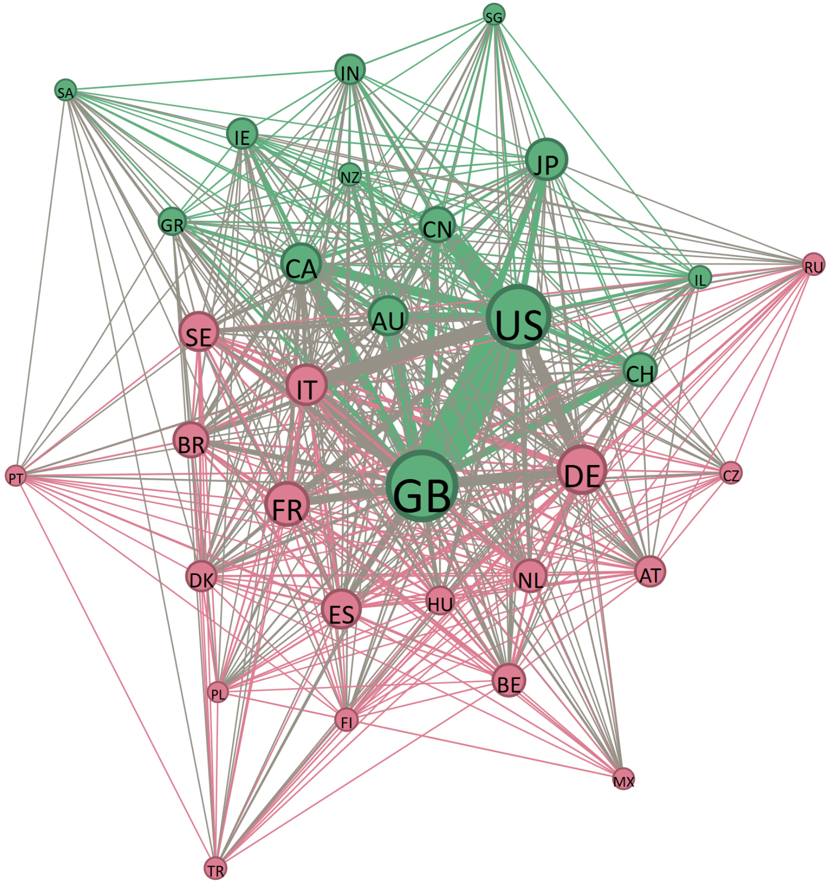


Figure 19: Top countries network (2 communities detected)

Concepts and keywords

HSDSLab
Human & Social Data Science

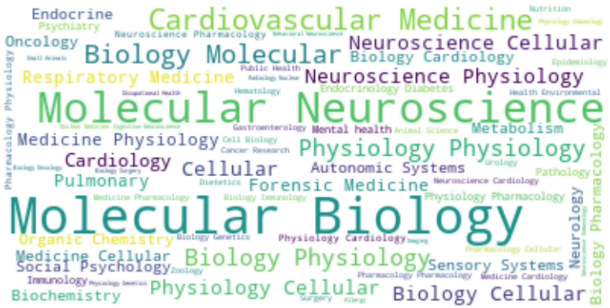
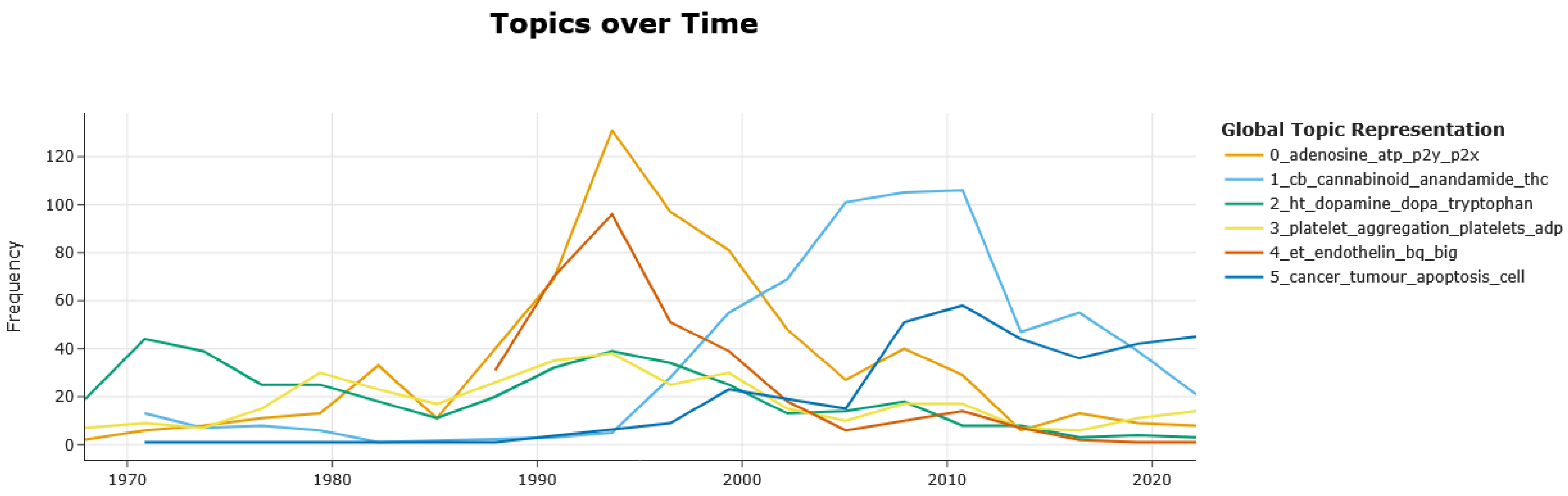


Figure 20: Subfields

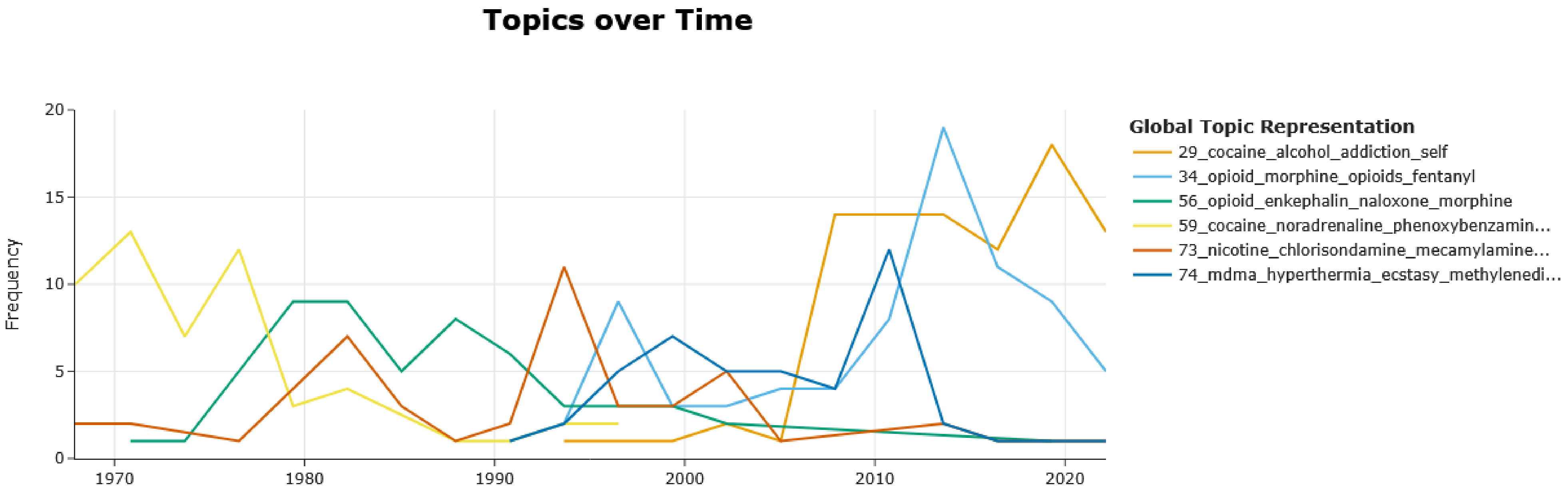
Figure 21: Topics

Figure 22: Topics since 2020, cleaned

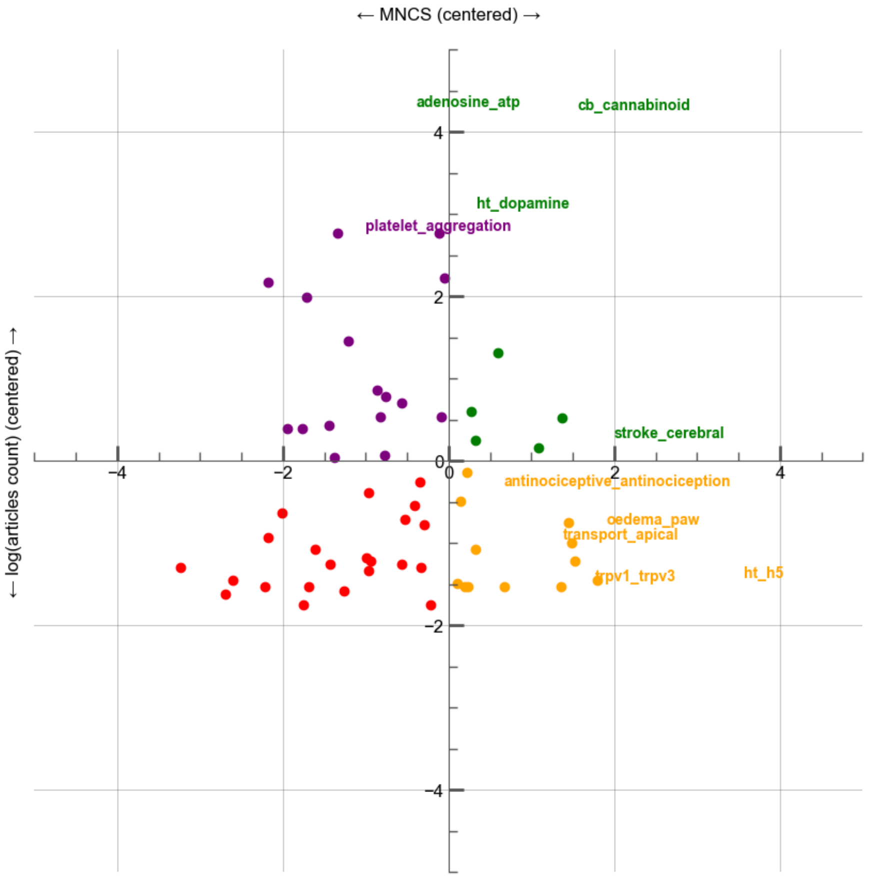
Concept and keywords



Concept and keywords



Link between topic and impact (MNCS)



Current limitations and perspectives

- **Very limited data before 1970, and even post-1970 the dataset remains incomplete**.
  - A more complete archive, including full metadata for all published works would allow better analysis.
- **Possible directions for future research:**
  - Partial modeling of impact to enable prediction for newly published articles.
  - More in-depth topic analysis.